EX-02-*Cross-Platform-Prompting-EvaluatingDiverse-Techniques-in-AI-Powered-TextSummarization*

AIM :

To evaluate and compare the effectiveness of prompting techniques (zero-shot, few-shot, chain-of-thought, role-based) across different AI platforms (e.g., ChatGPT, Gemini, Claude, Copilot) in a specific task: text summarization.

SCENARIO:

You are part of a content curation team for an educational platform that delivers quick summaries of research papers to undergraduate students. Your task is to summarize a 500-word technical article on "The Basics of Blockchain Technology" using multiple AI platforms and prompting strategies.

Your goal is to determine which combination of prompting technique + platform provides the best summary in terms of:

Accuracy

Coherence

Simplicity

Speed

User experience

Output

ZERO SHOT PROMPT

PROMPT: write a well defined 200words article on "The Basics of Block Chain Technology" with key points

CHATGPT:

The Basics of Blockchain Technology

Blockchain technology is a decentralized digital system used to record and store data securely across multiple computers. Instead of storing information in a single location, blockchain distributes the data across a network, making it more transparent, secure, and resistant to tampering. It was first introduced as the underlying technology for Bitcoin, a cryptocurrency, but today it is used in many fields such as finance, healthcare, and supply chain management.

A blockchain consists of a series of blocks that store data. Each block contains transaction information, a timestamp, and a unique cryptographic hash that links it to the previous block, forming a chain. This structure ensures that once data is recorded, it cannot be easily altered without changing all the following blocks in the chain.

Blockchain operates through a peer-to-peer network where participants validate transactions using consensus mechanisms. This eliminates the need for a central authority or intermediary, reducing costs and increasing efficiency.

Key Points:

- Decentralized system with no central authority
- Data stored in blocks linked together in a chain

- Highly secure due to cryptography
- Transparent and tamper-resistant
- Used in cryptocurrencies, smart contracts, and digital transactions

Overall, blockchain technology improves trust, security, and efficiency in digital record-keeping and transactions. 🔗 💻

GEMINI:

The Basics of Blockchain Technology

At its core, **blockchain** is a decentralized, distributed ledger technology that records transactions across many computers. Unlike a traditional database managed by a single entity (like a bank), blockchain ensures that no single person or organization has total control, making the data transparent and highly secure.

How It Works

Every transaction is bundled into a "block." Once a block is filled, it is encrypted and linked to the previous block using a unique digital fingerprint called a **hash**. This creates a chronological "chain." If anyone attempts to alter a past transaction, the hash changes, breaking the chain and immediately alerting the entire network.

Key Components

- **Decentralization:** Data is stored across a global network of "nodes" (computers), eliminating a single point of failure.

- **Immutability:** Once data is written, it cannot be deleted or changed without the consensus of the network.
- **Consensus Mechanisms:** Rules (like Proof of Stake) that nodes follow to agree that a transaction is valid.
- **Transparency:** In public blockchains, anyone can view the transaction history, though identities remain encrypted.

Beyond Cryptocurrency

While famous for powering Bitcoin, blockchain's potential in 2026 extends to **Smart Contracts** (self-executing agreements), supply chain tracking, and the tokenization of real-world assets like real estate. By removing intermediaries, it reduces costs and increases trust in the digital economy.

CLAUDE 2:

The Basics of Blockchain Technology

Blockchain technology is revolutionizing how we store and transfer data by creating secure, transparent, and decentralized digital ledgers. At its core, a blockchain is a chain of blocks containing transaction records that are linked together using cryptography. Key Points:

Decentralization: Unlike traditional databases controlled by single entities, blockchains operate across networks of computers, eliminating central authority and reducing vulnerability to manipulation.

Immutability: Once data is recorded in a block and added to the chain, it becomes nearly impossible to alter. Each block contains a cryptographic hash of the previous block, creating an unbreakable link.

Transparency: All network participants can view the entire transaction history, ensuring accountability while maintaining user privacy through pseudonymous addresses.

Consensus Mechanisms: Networks use protocols like Proof of Work or Proof of Stake to validate transactions and maintain agreement across all nodes without requiring trust between parties.

Security: Cryptographic techniques protect data integrity, making blockchain resistant to hacking and fraud.

Originally developed for Bitcoin in 2008, blockchain now extends beyond cryptocurrency into supply chain management, healthcare records, voting systems, and smart contracts. This technology promises to transform industries by enabling trust less transactions, reducing intermediaries, and creating more efficient systems for managing digital assets and information.

COPILOT:

The Basics of Blockchain Technology

Blockchain technology is a revolutionary system that enables secure, transparent, and decentralized record-keeping. At its core, a blockchain is a digital ledger that stores data in blocks, which are linked together in chronological order to form a chain. This structure ensures that once information is recorded, it cannot be altered without consensus, making blockchain highly trustworthy.

Key Points:

- **Decentralization**: Unlike traditional databases controlled by a single authority, blockchain operates across a network of computers, reducing risks of manipulation.
- **Transparency**: Every participant in the network can view transactions, ensuring accountability.
- **Security**: Data is encrypted and validated through consensus mechanisms like Proof of Work or Proof of Stake.
- **Immutability**: Once a block is added, it cannot be changed, preserving the integrity of records.
- **Applications**: Beyond cryptocurrencies like Bitcoin, blockchain is used in supply chain management, healthcare, voting systems, and smart contracts

FEW SHOT PROMPT

Prompt: **Example1:** Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence. These tasks include learning from data, recognizing patterns, understanding language, and making decisions. AI systems use techniques such as machine learning, deep learning, and natural language processing to improve their performance over time. AI is widely used in everyday applications like voice assistants, recommendation systems, healthcare diagnostics, and autonomous vehicles. By increasing efficiency and accuracy, AI helps solve complex problems and supports innovation across industries, transforming how people work, learn, and interact with technology.

Example2: Cyber Security is the practice of protecting computers, networks, systems, and data from digital attacks, unauthorized access, and damage. It involves technologies, processes, and policies designed to safeguard information from threats such as malware, phishing, ransomware, and hacking. Cyber security ensures the confidentiality, integrity, and availability of data. It plays a crucial role in protecting personal information, business operations, and national infrastructure. With the rapid growth of the internet and digital services, strong cyber security measures are essential to prevent data breaches, financial losses, and cybercrime, ensuring a safe and trusted digital environment.

Now Generate a summary of 200 words on The Basics of Basics of Block Chain Technology

CHATGPT:

Blockchain technology is a decentralized digital system used to securely record and manage transactions across a network of computers. Instead of relying on a central authority such as a bank or organization, blockchain allows multiple participants in a network to maintain and verify the same data. This distributed structure increases transparency, security, and trust among users.

A blockchain is made up of a series of blocks that store information such as transaction details, timestamps, and a cryptographic hash. Each block is connected to the previous one, forming a continuous chain of records. Once information is added to a block and verified by the network, it becomes extremely difficult to modify or delete, ensuring data integrity.

Blockchain operates using peer-to-peer technology and consensus mechanisms, which allow network participants to agree on the validity of transactions. This process removes the need for intermediaries and reduces the risk of fraud or manipulation. Cryptography is used to protect the data and ensure that only authorized transactions are recorded.

Blockchain technology is widely used in cryptocurrencies such as Bitcoin, but its applications extend to areas like supply chain management, digital identity verification, healthcare records, and smart contracts. By providing a secure, transparent, and efficient way to store and share data, blockchain technology is transforming many industries and improving the reliability of digital transactions. 🔗💻

GEMINI:

The Basics of Blockchain Technology

Blockchain is a revolutionary decentralized ledger technology designed to record transactions securely across a vast network of computers. Unlike traditional centralized systems, such as a bank's private database, blockchain operates on a peer-to-peer basis, ensuring that no single authority maintains absolute control over the data.

How the Process Works

The technology functions by grouping data into "blocks." Each block contains a cryptographic hash of the previous block, a timestamp, and transaction data. This structure creates a chronological, unbreakable chain. Because each block is mathematically linked to its predecessor, any attempt to alter

past information would require changing every subsequent block, making the system virtually immutable.

Core Pillars

- Decentralization: Data is distributed across "nodes," preventing a single point of failure.
- Transparency: All participants can view the transaction history, fostering high levels of trust.
- Consensus: The network uses protocols like Proof of Work or Proof of Stake to validate transactions without needing a middleman.

Beyond digital currencies, blockchain is now a foundational tool for smart contracts, secure voting, and supply chain transparency. By replacing blind trust with mathematical certainty, it provides a robust framework for the modern digital economy.

CLAUDE 2:

The Basics of Blockchain Technology Blockchain technology is a revolutionary digital system that enables secure, transparent, and decentralized recording of transactions and data. At its foundation, a blockchain is a distributed ledger consisting of blocks that contain transaction information, timestamps, and cryptographic links to previous blocks, forming an unchangeable chain. Unlike traditional centralized databases controlled by single organizations, blockchain operates across a network of computers called nodes, where each participant maintains a complete copy of the ledger. This decentralization eliminates the need for intermediaries and reduces the risk of data manipulation or system failure.

The technology employs cryptographic hashing to ensure data integrity, making it virtually impossible to alter past records without detection. Consensus mechanisms like Proof of Work or Proof of Stake validate new transactions, ensuring all network participants agree on the ledger's current state. Blockchain provides transparency by allowing authorized users to view transaction histories while maintaining privacy through pseudonymous addresses.

Originally created to support Bitcoin in 2008, blockchain has expanded far beyond cryptocurrency. Today, it powers applications in supply chain tracking, healthcare records management, digital identity verification, smart contracts, and voting systems. By enabling trustless peer-to-peer transactions, reducing operational costs, and increasing security, blockchain technology is transforming industries and reshaping how organizations manage data, conduct business, and establish trust in the digital economy.

COPILOT:

The Basics of Blockchain Technology

Blockchain technology is a digital innovation designed to securely record and share information without relying on a central authority. At its foundation, blockchain is a distributed ledger system where data is stored in blocks. Each block is linked to the previous one, forming a chain that is resistant to tampering. This structure ensures transparency, security, and trust among participants in the network.

Blockchain operates on the principle of decentralization, meaning that no single entity controls the data. Instead, multiple computers (nodes) maintain and verify the ledger, reducing risks of fraud or manipulation. Transactions are validated through consensus mechanisms such as Proof of Work or Proof of Stake, which guarantee accuracy and reliability. Once a block is added, it becomes immutable, meaning the information cannot be altered, preserving the integrity of records.

The technology also emphasizes transparency, as all participants can view transactions, fostering accountability. Beyond its initial use in cryptocurrencies like Bitcoin, blockchain has expanded into diverse fields including supply chain management, healthcare, finance, and smart contracts, where automated agreements execute without intermediaries.

By combining security, efficiency, and trust, blockchain is transforming industries and redefining how data is stored, shared, and utilized in the digital age.

CHAIN OF THOUGHT PROMPT

Prompt: summarize the topic The basics of block chain technology in 200 words in step by step

CHATGPT:

The Basics of Blockchain Technology (Step-by-Step Summary)

Step 1: Introduction
Blockchain technology is a decentralized digital system used to store and record information securely. It allows data to be

shared across a network of computers without the need for a central authority such as a bank or organization.

Step 2: Structure of Blockchain
Blockchain consists of a series of blocks that store data. Each block contains transaction details, a timestamp, and a unique cryptographic hash. These blocks are connected to each other, forming a continuous chain.

Step 3: Data Security
Once a block is added to the blockchain, the information cannot be easily changed. Cryptography protects the data and ensures that the records remain secure and tamper-resistant.

Step 4: Decentralized Network
Blockchain operates through a peer-to-peer network where many computers (called nodes) maintain and verify the same information. This distributed system increases transparency and reliability.

Step 5: Transaction Verification
Before a transaction is added to the blockchain, network participants verify it using consensus mechanisms. This ensures that only valid transactions are recorded.

Step 6: Applications
Blockchain is widely used in cryptocurrencies like Bitcoin. It is also used in supply chain management, digital identity systems, healthcare records, and smart contracts.

Overall, blockchain technology provides a secure, transparent, and efficient way to record and manage digital transactions.
🔗💻

GEMINI:

The Basics of Blockchain Technology: A Step-by-Step Summary

Blockchain is a decentralized, digital ledger technology that records transactions across a network of computers. Unlike a traditional bank, it operates without a central authority, ensuring transparency and security through a structured, multi-step process.

Step-by-Step Process

1. Transaction Request: A user initiates a transaction (e.g., sending digital currency or data).
2. Verification: The request is broadcast to a peer-to-peer network consisting of computers known as nodes.
3. Consensus: The nodes validate the transaction using algorithms (Consensus Mechanisms). This ensures the sender has the necessary assets and the transaction is legitimate.
4. Block Creation: Once verified, the transaction is combined with others to form a new block of data.
5. Hashing and Linking: The new block is assigned a unique cryptographic hash and linked to the hash of the previous block.
6. Finalization: The block is added to the existing chain in a permanent, chronological order. The ledger is updated across the entire network.

Key Benefits

- Immutability: Once a block is added, it is nearly impossible to alter, protecting data integrity.
- Security: Distributed storage means there is no single point of failure for hackers to exploit.
- Efficiency: By removing intermediaries like brokers or banks, blockchain reduces costs and speeds up global transactions.

CLAUDE 2:

The Basics of Blockchain Technology: Step-by-Step Step 1: Understanding the Structure Blockchain is a digital ledger composed of individual "blocks" that store transaction data. Each block contains information, a timestamp, and a unique cryptographic hash linking it to the previous block, forming a continuous chain.

Step 2: Decentralization Process Instead of storing data in one central location, blockchain distributes copies across multiple computers (nodes) in a network. Each participant maintains an identical version of the entire ledger.

Step 3: Transaction Initiation When someone initiates a transaction, it is broadcast to all nodes in the network for verification.

Step 4: Validation Through Consensus Network participants use consensus mechanisms like Proof of Work or Proof of Stake to validate the transaction's authenticity, ensuring no fraud occurs.

Step 5: Block Creation Once verified, the transaction is grouped with others into a new block, which receives its unique cryptographic hash.

Step 6: Adding to the Chain The new block is permanently added to the existing blockchain, creating an immutable record that cannot be altered without network consensus.

Step 7: Real-World Applications This technology now powers cryptocurrency, supply chain tracking, healthcare records, smart contracts, and digital identity systems, revolutionizing how we secure and share information across industries.

COPILOT:

Here's a clear step-by-step 200-word summary of *The Basics of Blockchain Technology*:

Step 1: Definition

Blockchain is a digital ledger system that records transactions in blocks. These blocks are linked together in sequence, forming a secure chain of data.

Step 2: Structure

Each block contains transaction details, a timestamp, and a cryptographic hash. Once added, the block is connected to the previous one, making the chain tamper-resistant.

Step 3: Decentralization

Unlike traditional databases controlled by a single authority, blockchain operates across a distributed network of computers (nodes). This ensures no single point of failure.

Step 4: Transparency

All participants in the network can view transactions, promoting accountability and trust.

Step 5: Security

Blockchain uses encryption and consensus mechanisms such as Proof of Work or Proof of Stake to validate transactions, ensuring accuracy and preventing fraud.

Step 6: Immutability

Once data is recorded, it cannot be altered without network consensus, preserving the integrity of information.

Step 7: Applications

Beyond cryptocurrencies like Bitcoin, blockchain is used in supply chain tracking, healthcare records, financial services, voting systems, and smart contracts.

Step 8: Impact

By combining decentralization, transparency, and security, blockchain is transforming industries, offering efficient, trustworthy, and innovative ways to manage digital information.

ROLE BASED PROMPT

Prompt: Pretend you are a block chain expert and give me an article on "The basics of block chain Technology" in 150 words

CHATGPT:

As a blockchain expert, I can explain that blockchain technology is a decentralized digital ledger used to securely record and verify transactions across a network of computers.

Unlike traditional databases controlled by a central authority, blockchain distributes data across many nodes, ensuring transparency and reducing the risk of fraud or manipulation.

A blockchain is made up of blocks that store transaction details, timestamps, and a cryptographic hash that links each block to the previous one. This structure creates a secure chain of records where once data is added, it becomes extremely difficult to alter. The system relies on cryptographic techniques and consensus mechanisms to validate and approve transactions within the network.

Blockchain technology was first introduced with cryptocurrencies like Bitcoin, but its applications now extend far beyond digital currency. It is used in areas such as supply chain management, digital identity verification, healthcare records, and smart contracts. By providing security, transparency, and decentralization, blockchain technology is transforming the way digital transactions and data management are handled across industries. 🔗 💻

GEMINI:

The Expert's Guide: Basics of Blockchain Technology

As a blockchain architect, I define blockchain as a decentralized, distributed ledger that records transactions across a global peer-to-peer network. Its primary purpose is to establish trustless security; it removes the need for intermediaries like banks by using math and consensus instead of centralized authority.

How the Architecture Functions

Data is bundled into blocks. Each block contains a unique cryptographic hash and the hash of the block preceding it. This creates a mathematical "chain." If a single digit is altered in an old block, its hash changes, breaking the entire chain and alerting the network.

Three Pillars of the Technology

- Immutability: Once data is "chained," it cannot be deleted or reversed.
- Transparency: Every node holds a copy of the ledger, making fraud nearly impossible to hide.
- Consensus: Protocols like Proof of Stake ensure all nodes agree on the ledger's state.

Beyond Bitcoin, we use this for Smart Contracts—self-executing code that automates legal and financial agreements with zero human error.

CLAUDE 2:

The Basics of Blockchain Technology As a blockchain expert, I can tell you that blockchain is essentially a revolutionary distributed ledger system that fundamentally changes how we handle digital trust and transactions.

Think of blockchain as a digital notebook that's simultaneously copied across thousands of computers worldwide. When a transaction occurs, it's grouped into a "block" with other transactions. This block is then cryptographically sealed and

linked to the previous block, creating an unbreakable chain of records.

What makes blockchain powerful is its consensus mechanism. Before any block joins the chain, the majority of network participants must validate it, eliminating the need for banks or intermediaries. Once recorded, data becomes virtually immutable—altering it would require controlling over 51% of the network, which is practically impossible in large blockchains.

Beyond Bitcoin, I've seen blockchain transform supply chains, enable self-executing smart contracts, secure medical records, and create transparent voting systems. The technology's ability to provide transparency, security, and decentralization simultaneously makes it a game-changer for the digital economy.

COPILOT:

The Basics of Blockchain Technology

Blockchain technology is a decentralized digital ledger system designed to securely record and share information. Instead of relying on a central authority, blockchain operates across a network of computers, known as nodes, which collectively validate and store transactions. Data is grouped into blocks, each containing transaction details, a timestamp, and a cryptographic hash. These blocks are linked together in

chronological order, forming an immutable chain that prevents tampering.

One of the core strengths of blockchain is decentralization, which eliminates single points of failure and reduces risks of fraud. It also ensures transparency, as all participants can view transactions, fostering trust. Security is maintained through consensus mechanisms such as Proof of Work or Proof of Stake, which verify the accuracy of data before it is added to the chain.

Beyond cryptocurrencies like Bitcoin, blockchain has diverse applications in supply chain management, healthcare, finance, and smart contracts. By combining transparency, security, and efficiency, blockchain is transforming industries and redefining how digital information is stored, shared, and trusted.

Evaluation Criteria

- Accuracy – correctness of blockchain concepts
- Coherence – logical flow and organization
- Simplicity – ease of understanding
- Speed – how quickly a usable summary is produced (minimal refinement needed)
- User Experience – readability, clarity, and alignment with user intent

Overall Evaluation AI Model Performance Level :

ChatGPT ⭐⭐⭐⭐⭐ (Best overall adaptability)

Claude 2 ⭐⭐⭐⭐☆ (Best depth & narrative quality)
Copilot ⭐⭐⭐⭐☆ (Best structured technical output)
Gemini ⭐⭐⭐☆ (Best for basic summaries)

RESULT: Therefore,to evaluate and compare the effectiveness of prompting techniques (zero-shot, few-shot, chain-of-thought, role-based) across different AI platforms (e.g., ChatGPT, Gemini, Claude, Copilot) in a specific task: text summarization is completed successfully.